KULBHUSHAN CHAND

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Emerging Technologies Lab
Department of Electronics and Communication Engineering
Dr B R Ambedkar National Institute of Technology
(https://www.nitj.ac.in/)
Jalandhar – 144011, Punjab, India

EDUCATION

PhD | Electronics and Communication Engineering **2015 - Present**
Dr B R Ambedkar National Institute of Technology, Jalandhar, India
Dissertation: Low cost multimodal biofeedback for mental relaxation
Advisor: Prof. Arun Khosla

MTech | Electronics and Communication Engineering **2011 - 2013**
Lovely professional University, Phagwara, India
Thesis: A new approach for ECG signal denoising using Adaptive Filtering

BTech | Electronics and Communication Engineering **2006 - 2010**
Guru Nanak Dev University, Regional Campus, Jalandhar, India
Graduated with Hons.

WORK EXPERIENCE

Research Scholar **Aug 2015 - Present**
Department of ECE, Dr B R Ambedkar National Institute of Technology

Assistant Professor **Aug 2013 – July 2015**
Department of ECE, Lovely Professional University

Lecturer **Dec 2010 – May 2011**
Department of ECE, Lovely Professional University

HONORS AND AWARDS

UGC NET with JRF **2013**

University Topper (MTech) **2013**

Honors Roll (BTech) **2010**

RESEARCH EXPERIENCE

Project Team Member **Feb 2020 – Feb 2022**
ISRO Space Technology Incubation Center (S-TIC),
Dr B R Ambedkar National Institute of Technology
Project - Strengthening Ridge to Valley based NRM planning under Mission Water Conservation using Geospatially enabled Serious Gaming

- Developing gamification framework for the project

Doctoral Researcher **2015 - Present**
Department of ECE, Dr B R Ambedkar National Institute of Technology
Advisor: Prof. Arun Khosla

- Developed various H/W and S/W solutions as a part of "low cost biofeedback system for mental relaxation"

TEACHING EXPERIENCE

Teaching Assistant **Aug 2015 to Jul 2020**
Department of ECE, Dr B R Ambedkar National Institute of Technology

- Co-created and introduced Gamification, an undergraduate course averaging 120 students per semester.
- Designed various pedagogy, tools, assignments, and exams.
- Other courses taught includes Network Analysis and Synthesis, Electronic Devices and Simulation Lab, Basic Electronics Lab.

Assistant Professor **Aug 2013 to Jun 2015**
Department of ECE, Lovely Professional University

- Courses and labs taught includes Basic Electrical Engineering, Electronic Devices and Circuits, Probability and Stochastic Theory, Signal and Systems, Digital Signal Processing, Adaptive Filter Theory, Basic Electrical lab, Electronic Devices and Circuits lab, Digital Signal Processing lab.
- Developed quizzes, exams, assignments and homework.
- Revised the syllabus an instruction plans to meet accreditation standards.
- Guided undergraduate students for their major project.

Teaching Assistant **Jan 2012 to Nov 2012**
Department of ECE, Lovely Professional University

- Conducted lab Basic Electronics Lab course for undergraduate students.
- Evaluated practical, viva-voce, and written exam.

Lecturer **Dec 2010 to May 2011**
Department of ECE, Lovely Professional University

- Taught Microcontroller and its Applications course to undergraduate students and conducted Microcontroller and Microprocessor lab.
- Developed exams, assignments and homework.
- Guided undergraduate students for their major project.

PUBLICATIONS

Book Chapters

Jain, B., Khosla, A., **Chand, K.**, & Ahuja, K. (2019). The Pursuit of Zero Food Waste: A Gamified Approach Promoting Avoidance of Dormitory Mess Food Wastage in Educational Institutions. In Global Initiatives for Waste Reduction and Cutting Food Loss (pp. 243–267). IGI Global. https://doi.org/10.4018/978-1-5225-7706-5.ch013

Mukherjee, A., Goyal, P., Singh, A., Khosla, A. K., Ahuja, K., & **Chand, K.** (2019). Pursuit of Research: A Gamified Approach Promoting Research Engagement Among Undergraduate Students. In Fostering Multiple Levels of Engagement in Higher Education Environments (pp. 132–155). IGI Global. https://doi.org/10.4018/978-1-5225-7470-5.ch006

Padmakar, G. S. V., Khosla, A., & **Chand, K.** (2019). LearnEasy-Android Application as a Technological Intervention for Children With Dyslexia. In Emerging Trends in the Diagnosis and Intervention of Neurodevelopmental Disorders (pp. 236–248). IGI Global. https://doi.org/10.4018/978-1-5225-7004-2.ch012

Setia, D., Singh, R., Sharma, A., Khosla, A., Ahuja, K., & **Chand, K.** (2019). Enhancing Tourism and Cultural Experience Through Gamification. In Positioning and Branding Tourism Destinations for Global Competitiveness (pp. 152–171). IGI Global. https://doi.org/10.4018/978-1-5225-7253-4.ch007

TALKS

Invited Talks

Octalysis — Nov 2020
AICTE ATAL Sponsored Online Faculty Development Program on Gamification | Dr B R Ambedkar National Institute of Technology, Jalandhar

Arduino and Open-Source Technologies — Jun 2019
One day workshop | DAV Institute of Engineering and Technology, Jalandhar

PROFESSIONAL DEVELOPMENT

Short Term Courses (Attended)

Biology for Engineers | *Indian Institute of Technology Bombay* — Jun 2016

Brain Science & Technology: Tools and Techniques to understand the Human Brain | *Indian Institute of Technology Delhi* — Dec 2016

Autism and Assisted Technologies (Gian Course) | *Dr B R Ambedkar National Institute of Technology* — Sep 2020

Computational Cardiac Engineering - Predictive Modelling and Innovative Sensing Systems | *Rajasthan Technical University* — Sep 2020

Machine Learning Using Python | *Dr B R Ambedkar National Institute of Technology* — Sep 2020

The Education Factory | *Dr B R Ambedkar National Institute of Technology* — Sep 2020

Certifications

The Octalysis Framework - Level 1 | *The Octalysis Group* — Dec 2017

MOOC Courses (with certificates)

Statistics in Medicine | *Stanford University* — Mar 2020

Computer Science 101 | *Stanford University* — Mar 2020

Introduction to CAD: Learn Fusion 360 in 90 minutes | *Autodesk* — Sep 2019

Vital Signs: Understanding What the Body Is Telling Us | *University of Pennsylvania* — May 2018

Advanced MATLAB for Scientific Computing | *Stanford University* — Dec 2017

Introduction to Game Development | *Michigan State University* — Mar 2017

Calculus One | *The Ohio State University* — May 2013

Image and video processing | *Duke University* — Mar 2013

COMMUNITY OUTREACH

Arduino workshop for kids | *Jalandhar* Nov 2017

SKILLS

Programming: Assembly, C, C++, R, Lua, MATLAB
Applications: NI Multisim, Autodesk Fusion 360, KiCad, Unity3D, RStudio, JASP
Operating Systems: Windows
Others: Octalysis-Level1

PROJECTS

Software Projects

AfDaq: A plug and play MATLAB based tool for biofeedback and arduino based instruments

BioNES: A plug-and-play MATLAB based tool to use NES games for multimodal biofeedback

MCP4251: Arduino library for MCP4251 Digital Potentiometer

classroom-gamification: Tool to gamify your classroom with an online leaderboard

Book Projects

GamificationCourseBook: An online coursebook for Gamification (ECX-488)

ArduinoWorkshopForKids: Fun experiments with arduino for kids

Research Projects

AfDaq: An Open-source MATLAB based Real-time Multi-channel Data Acquisition Tool for Biofeedback and Arduino based Instruments

LANGUAGES

English: Advanced
Hindi: Native Language
Punjabi: Native Language

PERSONAL

Birth Year: 1988
Citizenship: Indian
Residence: India (Permanent residence)

REFERENCES

Prof. Arun Khosla
Emerging Technologies Lab
Department of Electronics and Communication Engineering
Dr B R Ambedkar National Institute of Technology (http://www.nitj.ac.in/)

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